

Sentiment Analysis: VADER

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Text Processing

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Learning objectives

- ▶ Explain how SA lexicons are created
- ▶ Explain the VADER model
- ▶ Review your knowledge about lexicon-based SA

Sentiment Analysis: lexicon-based

How to obtain lexica of emotion words?

1. Collect relevant words/phrases expressing sentiment
 2. Determine the emotion of these subjective texts
- Can be done **manually**
- ▶ word list with pre-assigned emotional weights
- Can be done **semi-automatically**
- ▶ require a dictionary of **seed emotion words**
 - ▶ **dictionary-based**
 - ▶ find synonyms/antonyms using linguistic resources like e.g. WordNet
 - <https://wordnet.princeton.edu> and <http://projects.illc.uva.nl/EuroWordNet>
 - <http://globalwordnet.org/resources/wordnets-in-the-world/>
 - ▶ **corpus-based**
 - ▶ find synonyms/antonyms in corpora

Sentiment Analysis: lexicon-based

What do lexica of emotion words contain?

▶ Adjectives

- ▶ **positive**: honest, important, mature, large, patient, ...
- ▶ **negative**: harmful, hypocritical, inefficient, insecure, ...

▶ Verbs

- ▶ **positive**: praise, love, ...
- ▶ **negative**: blame, criticize, ...

▶ Nouns

- ▶ **positive**: pleasure, enjoyment, ...
- ▶ **negative**: pain, criticism, ...

▶ Phrases: for **collocations**, also an alternative to **intensifiers**

- ▶ **positive**: "very efficient", "low cost", ...
- ▶ **negative**: "many problems", "lot of bugs", ...

Sentiment Analysis: lexicon based: manually created resources

Manually created resources by experts

- ▶ **SentiWordNet** database:
<http://ontotext.fbk.eu/sentiwn.html>
 - ▶ **Wordnet**: words grouped in sets of synonyms (**synsets**)
 - ▶ with semantic relations between them: synonyms, antonyms, hypernyms, etc.
 - sentiment score added **negative**, **positive** and **neutral**
- ▶ **Linguistic Inquiry and Word Count (LIWC) lexicon**
 - ▶ made by psychologists
 - ▶ words with several **emotional and other dimensions**
- ▶ **General Inquirer**
 - ▶ terms with various types of **negative** or **positive** semantic orientation
 - <http://www.wjh.harvard.edu/~inquirer/homecat.htm>

SentiWordNet



SentiWordNet

ADJECTIVE

terrific1 01513619

very great or intense; "a terrific noise"; "a terrific thunderstorm storm"; "fought a terrific battle"

Feedback on SentiWordNet values: [They are OK.](#) [Suggest your values.](#)

wondrous#1 wonderful#1 tremendous#2 **terrific2** rattling#1 marvelous#1 marvelous#1 howling#1 grand#4 fantastic#2 01676517

extraordinarily good or great ; used especially as intensifiers; "a fantastic trip to the Orient"; "the film was fantastic!"; "a howling success"; "a marvelous collection of rare books"; "had a rattling conversation about politics"; "a tremendous achievement"

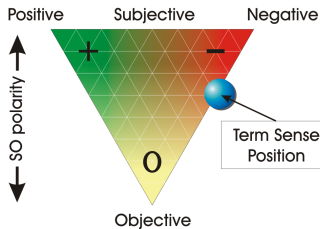
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terrifying#1 **terrific3** 00196449

causing extreme terror; "a terrifying wall"

Feedback on SentiWordNet values: [They are OK.](#) [Suggest your values.](#)

← PN polarity →



Linguistic Inquiry and Word Count (LIWC) lexicon

Table 1. LIWC2007 Content Categories of Affective, Cognitive and Perceptual Processes, with Subcategories, Abbreviation and Examples (Pennebaker, Booth, & Francis, 2007).

Category	Abbreviation	Examples
Affective processes	affect	Happy, cried, abandon
Positive emotion	posemo	Love, nice, sweet
Negative emotion	negemo	Hurt, ugly, nasty
Anxiety	anx	Worried, fearful, nervous
Anger	anger	Hate, kill, annoyed
Sadness	sad	Crying, grief, sad
Cognitive processes	cognitive	Cause, know, ought
Insight	insight	Think, know, consider
Causation	cause	Because, effect, hence
Perceptual processes	percept	Observing, heard, feeling
See	see	View, saw, seen
Hear	hear	Listen, hearing
Feel	feel	Feels, touch

[Brönnimann et al., 2013]

Creating / expanding a lexicon → semi-automatic methods

Semi-automatically created from **seed words**

- ▶ start with **seed negative** and **positive** words
 - ▶ **dictionary-based**: synonyms/antonyms in dictionaries like WordNet
 - ▶ **corpus-based**: search on large corpora, like the web, for **patterns**
 - ▶ "beautiful and" (+)
 - ▶ "low cost but" (-)
 - ▶ "very nice and" (+)

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Very nice and **perfect**

★★★★★ Note : 5 - Critique de a TripAdvisor user - Prix : \$ (Based on Average Nightly Rates for a Standard Room from our Partners)

Very nice and **comfortable**

★★★★★ Note : 5 - Critique de a TripAdvisor user - Prix : \$ (Based on Average Nightly Rates for a Standard Room from our Partners)

Very nice and **quiet apartment near the beach**

★★★★★ Note : 4,7 - 6 avis

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Very expensive but **excellent food and wine**.

★★★★★ Note : 5 - Critique de a TripAdvisor user

Very expensive **but good food.**

★★★★★ Note : 3 - Critique de a TripAdvisor user

Very expensive but **disappointing..**

★★★★★ Note : 1 - Critique de a TripAdvisor user

Creating / expanding a lexicon → semi-automatic methods

If you have a corpus with assigned sentiment per sentence:

Doc	Words	Class
1	great movie, excellent plot, renowned actors	positive
2	I had not seen a fantastic plot like this in good 5 years. amazing!!!	positive
3	lovely plot, amazing cast, somehow I am in love with the bad guy	positive
4	bad movie with great cast, but very poor plot and unimaginative ending	negative
5	I hate this film, it has nothing original	negative
6	great movie, but not...	negative
7	very bad movie, I have no words to express how I dislike it	negative

Extract lists of words only from positive (negative) examples

- ▶ only some POS categories (e.g. adjectives)
- ▶ most frequent only

POSITIVE: great, excellent, renowned, fantastic, amazing, lovely, love

NEGATIVE: bad, poor, unimaginative, hate, dislike

[VADER] Valence Aware Dictionary for sEntiment Reasoning

VADER [Hutto and Gilbert, 2014]

- ▶ Uses **lexicon and grammatical / syntactic rules** in order to classify a text
- ▶ Specially designed for **Social Media / user generated content** (microblogs)
- ▶ Available on the NLTK toolkit for python
- ▶ One of the most used sentiment analysers

VADER [Hutto and Gilbert, 2014]

Objectives:

- ▶ optimised for **social media**, but **generalisable to other domains**

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- ▶ **fast**

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- ▶ **fast**
- ▶ **speed vs. performance trade-off** is not an issue

VADER [Hutto and Gilbert, 2014]

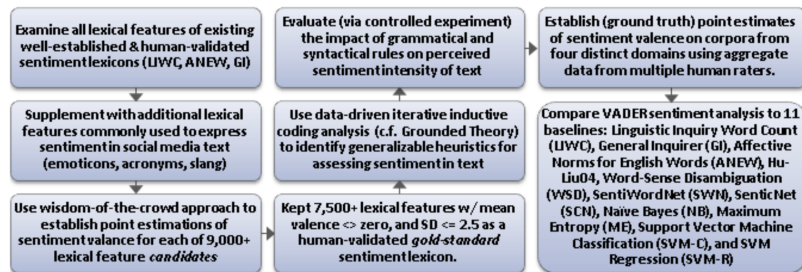


Figure 1: Methods and process approach overview.

VADER: lexicon creation

- ▶ Start with a **list** from LIWC, ANEW, GI

VADER: lexicon creation

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- ▶ Add lists of
 - ▶ Western-style **emoticons**
 - ▶ **sentiment-related acronyms** (e.g. LOL)
 - ▶ **slangs** (e.g. meh, nah)

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- ▶ Add lists of
 - ▶ Western-style **emoticons**
 - ▶ **sentiment-related acronyms** (e.g. LOL)
 - ▶ **slangs** (e.g. meh, nah)
- ▶ Ask humans to **annotate/validate the candidate features** ($\approx 9k$ candidates)

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"Wisdom of the crowd" → 10 independent raters per candidate feature (using AMT)

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"**Wisdom of the crowd**" → 10 independent raters per candidate feature (using AMT)

- ▶ Features are rated in a scale from **-4 (extremely negative)** to **4 (extremely positive)**
 - **neutral (0)** is allowed as well as "neither" or "N/A"

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- ▶ Features should have
 - ▶ a non-zero mean rating
 - ▶ standard deviation < 2.5 (10 raters)
- ▶ Selected features ($\approx 7.5k$) indicate **both sentiment polarity and intensity**
 - "okay": positive +0.9; "good": positive +1.9; "great": positive +3.1
 - "horrible": negative -2.5; frowning emoticon "(": negative -2.2

Lexicon available at:

https://comp.social.gatech.edu/papers/hutto_ICWSM_2014.tar.gz

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 - no more than 1 STDEV away from the mean

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- ▶ English language quality control
- ▶ Pre-task quality control (sentiment rating training)
- ▶ Task quality control (gold standard examples in the 25 examples batch)
 - no more than 1 STDEV away from the mean
- ▶ Extra payment for workers matching the group mean

VADER: rules

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- ▶ Manual analysis of 400 positive and 400 negative tweets
 - Tweets pre-classified by **Pattern.en sentiment engine**
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 - Humans assigned **-4 to 4 intensity scores**
- ▶ Qualitative analysis → **characteristic that impact perceived intensity**
- ▶ Five general rules:
 - ▶ Exclamation rule
 - ▶ Capitalisation rule
 - ▶ Degree modifiers (intensifiers)
 - ▶ **Contrastive conjunction "but"** → text following the conjunction being dominant
 - *The food here is great, **but the service is horrible***
 - ▶ Negation rule (considering trigrams preceding a lexicon feature)

VADER: rules

Impact of each rule is defined by using a gold standard manually annotated by humans

Test Condition	Example Text
Baseline	Yay. Another good phone interview.
Punctuation1	Yay! Another good phone interview!
Punctuation1 + Degree Mod.	Yay! Another extremely good phone interview!
Punctuation2	Yay!! Another good phone interview!!
Capitalization	YAY. Another GOOD phone interview.
Punct1 + Cap.	YAY! Another GOOD phone interview!
Punct2 + Cap.	YAY!! Another GOOD phone interview!!
Punct3 + Cap.	YAY!!! Another GOOD phone interview!!!
Punct3 + Cap. + Degree Mod.	YAY!!! Another EXTREMELY GOOD phone interview!!!

Table 2: Example of baseline text with eight test conditions comprised of grammatical and syntactical variations.

VADER: rules

Impact of each rule is defined by using a gold standard manually annotated by humans

Test Condition	<i>t</i>	<i>p</i>	Diff.	95% C.I.
Punctuation (. vs !)	19.02	< 2.2e-16	0.291	0.261 - 0.322
Punctuation (! vs !!)	16.53	2.7e-16	0.215	0.188 - 0.241
Punctuation (!! vs !!!)	14.07	1.7e-14	0.208	0.178 - 0.239
All CAPS (w/o vs w)	28.95	< 2.2e-16	0.733	0.682 - 0.784
Deg. Mod. (w/o vs w)	9.01	6.7e-10	0.293	0.227 - 0.360

Table 3: Statistics associated with grammatical and syntactical cues for expressing sentiment intensity. Differences in means were all statistically significant beyond the 0.001 level.

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Contrastive conjunction:

- ▶ divide the text before/after the conjunction "but"
- before "but" → diminished the intensity by 50%
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The movie is boring, but the actors are amazing!

The movie is boring = -2.3

the actors are amazing! = 3.5

Applying the contrastive conjunction rule:

The movie is boring = $-2.3 * 0.5 = -1.15$

the actors are amazing! = $3.5 * 1.5 = 5.25$

VADER: final result

Compound value: summed valence scores of each word that appears in the lexicon, adjusted according to the rules and normalised between **-1 (extremely negative)** and **1 (extremely positive)**

$$compound = \frac{score}{score^2 + \alpha}$$

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- compound ≥ 0.5 → **positive**
- compound ≤ -0.5 → **negative**
- **neutral** otherwise

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Positive (**pos**), negative (**neg**) and neutral (**neu**): ratio of proportions of words in the each category

VADER: evaluation

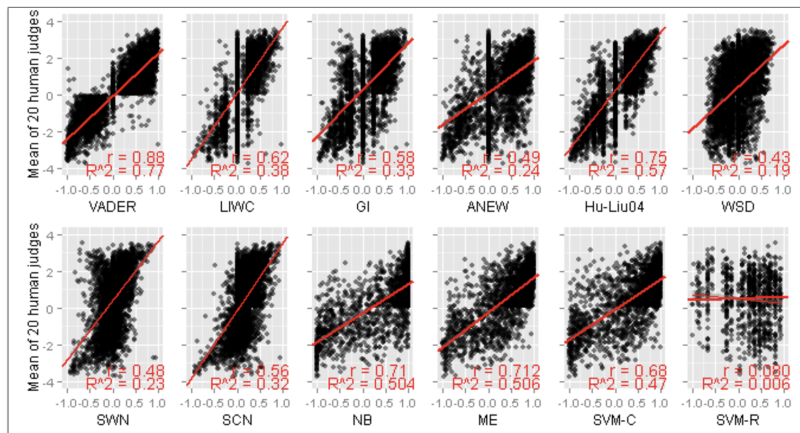


Figure 3: Sentiment scores from VADER and 11 other highly regarded sentiment analysis tools/techniques on a corpus of over 4K tweets. Although this figure specifically portrays correlation, it also helps to visually depict (and contrast) VADER's classification precision, recall, and F1 accuracy within this domain (see Table 4). Each subplot can be roughly considered as having four quadrants: true negatives (lower left), true positives (upper right), false negatives (upper left), and false positives (lower right).

VADER: evaluation

		Correlation to ground truth (mean of 20 human raters)	3-class (positive, negative, neutral) Classification Accuracy Metrics			Ordinal Rank (by F1)			Correlation to ground truth (mean of 20 human raters)	3-class (positive, negative, neutral) Classification Accuracy Metrics		
			Overall Precision	Overall Recall	Overall F1 score					Overall Precision	Overall Recall	Overall F1 score
Social Media Text (4,200 Tweets)												
Ind. Humans	0.888	0.95	0.76	0.84	2	1*	1	0.899	0.95	0.90	0.92	
VADER	0.881	0.99	0.94	0.96	3		2	0.451	0.70	0.55	0.61	
Hu-Liu04	0.756	0.94	0.66	0.77	3		3	0.416	0.66	0.56	0.59	
SCN	0.568	0.81	0.75	0.75	4		7	0.210	0.60	0.53	0.44	
GI	0.580	0.84	0.58	0.69	5		5	0.343	0.66	0.50	0.55	
SWN	0.488	0.75	0.62	0.67	6		4	0.251	0.60	0.55	0.57	
LIWC	0.622	0.94	0.48	0.63	7		9	0.152	0.61	0.22	0.31	
ANEW	0.492	0.83	0.48	0.60	8		8	0.156	0.57	0.36	0.40	
WSD	0.438	0.70	0.49	0.56	9		6	0.349	0.58	0.50	0.52	
Amazon.com Product Reviews (3,708 review snippets)												
Ind. Humans	0.911	0.94	0.80	0.85	1	1	1	0.745	0.87	0.55	0.65	
VADER	0.565	0.78	0.55	0.63	2		2	0.492	0.69	0.49	0.55	
Hu-Liu04	0.571	0.74	0.56	0.62	3		3	0.487	0.70	0.45	0.52	
SCN	0.316	0.64	0.60	0.51	7		7	0.252	0.62	0.47	0.38	
GI	0.385	0.67	0.49	0.55	5		5	0.362	0.65	0.44	0.49	
SWN	0.325	0.61	0.54	0.57	4		4	0.262	0.57	0.49	0.52	
LIWC	0.313	0.73	0.29	0.36	9		9	0.220	0.66	0.17	0.21	
ANEW	0.257	0.69	0.33	0.39	8		8	0.202	0.59	0.32	0.35	
WSD	0.324	0.60	0.51	0.55	6		6	0.218	0.55	0.45	0.47	
NY Times Editorials (5,190 article snippets)												

Table 4: VADER 3-class classification performance as compared to individual human raters and 7 established lexicon baselines across four distinct domain contexts (clockwise from upper left: tweets, movie reviews, product reviews, opinion news articles).

VADER: evaluation

	3-Class Classification Accuracy (F1 scores)			
	Test Sets			
	Tweets	Movie	Amazon	NYT
VADER	0.96	0.61	0.63	0.55
NB (tweets)	0.84	0.53	0.53	0.42
ME (tweets)	0.83	0.56	0.58	0.45
SVM-C (tweets)	0.83	0.56	0.55	0.46
SVM-R (tweets)	0.65	0.49	0.51	0.46
NB (movie)	0.56	0.75	0.49	0.44
ME (movie)	0.56	0.75	0.51	0.45
NB (amazon)	0.69	0.55	0.61	0.48
ME (amazon)	0.67	0.55	0.60	0.43
SVM-C (amazon)	0.64	0.55	0.58	0.42
SVM-R (amazon)	0.54	0.49	0.48	0.44
NB (nyt)	0.59	0.56	0.51	0.49
ME (nyt)	0.58	0.55	0.51	0.50

Table 5: Three-class accuracy (F1 scores) for each machine trained model (and the corpus it was trained on) as tested against every other domain context (SVM models for the movie and NYT data were too intensive for our multicore CPUs with 94GB RAM)

[Sentiment Analysis] Recap

Sentiment Analysis: Motivations + Concepts

Motivations

- ▶ Goal of Sentiment Analysis
- ▶ Examples of applications
- ▶ Sentiment analysis vs. opinion mining

Concepts

- ▶ Target of opinions
- ▶ Factual vs. subjective data

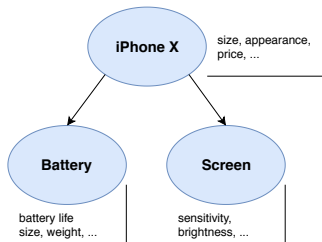
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Sentiment Analysis: Bing Liu's model

On 14th Nov. 2021, Billy S wrote:

“The new red iPhone is amazing. I bought mine three days ago and I am impressed. The touch screen works much better than my previous phone and the battery lasts for over 2 days. The camera is also very accurate and I can take nice pictures of my dog. The only drawback is the price: it is as expensive as a new laptop!”

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- ▶ o_j :
- ▶ f_{jk} :
- ▶ so_{ijkl} :
- ▶ opinion holder h_i :
- ▶ time t_l :

Sentiment Analysis: Bing Liu's model

On 20th Apr. 2022, Billy S wrote:

"My new bike trailer is amazing! It is large so can fit in comfortably. Mum said that it is also great because it is easy to assemble. She said it is also great that the wheels are easy to clean. I also love the colour, bright red is my favourite."

Sentiment Analysis: Bing Liu's model

On 20th Apr. 2022, Billy S wrote:

"My new bike trailer is amazing! It is large so can fit in comfortably. Mum said that it is also great because it is easy to assemble. She said it is also great that the wheels are easy to clean. I also love the colour, bright red is my favourite."

► o_j :

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- ▶ so_{ijkl} :
- ▶ opinion holder h_i :
- ▶ time t_l :

Sentiment Analysis: Granularity levels

- ▶ Document-level
- ▶ Sentence-level
- ▶ Feature-level

Sentiment Analysis: Lexicon-based approaches

- ▶ Binary
- ▶ Gradable / Graded / Weighted

Sentiment Analysis: Lexicon-based approaches → gradable

- ▶ Lexicon with polarity of words
- ▶ Six rules:
 - ▶ Negation rule
 - ▶ Capitalisation rule
 - ▶ Intensifier rule
 - ▶ Diminisher rule
 - ▶ Exclamation rule
 - ▶ Emoticon rule

Sentiment Analysis: Lexicon-based approaches → gradable

I really like the new movie with The Rock!

► Lexicon:

→ $\text{emotion}(\text{like}) = 2$

► Intensifier:

→ $\text{emotion}(\text{really}) = 2$

► Exclamation:

→ $\text{emotion}(!) = 1$

Final score: ?

Sentiment Analysis: Lexicon-based approaches → gradable

The Fast and Furious franchise is not great, it is boring and stupid
☹

► Lexicon:

→ $\text{emotion}(\text{great}) = 2$

→ $\text{emotion}(\text{boring}) = -2$

→ $\text{emotion}(\text{stupid}) = -3$

► Emoticon:

→ $\text{emotion}(\text{☹}) = -1$

Final score: ?

Reading material

- ▶ C. J. Hutto & Eric Gilbert (2014).
VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text
<https://www.aaai.org/ocs/index.php/ICWSM/ICWSM14/paper/viewPaper/8109>
- ▶ Daniel Jurafsky & James H. Martin (2021).
Chapter 20: Lexicons for Sentiment, Affect, and Connotation.
Speech and Language Processing, 3rd ed. draft.
<https://web.stanford.edu/~jurafsky/slp3/20.pdf>

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Do testimonies of traumatic events differ depending on the interviewer? .

Hutto, C. and Gilbert, E. (2014).

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In Proceedings of the Eighth International Conference on Weblogs and Social Media (ICWSM-14), pages 216–225, Ann Arbor, MI. AAAI.